PROJECT PROPOSAL FORM

2021 CHED-INITIATED INSTITUTIONAL DEVELOPMENT AND INNOVATION GRANT

I. PROJECT INFORMATION

HEI Proponent	Partido State University
Project Title	Establishment of Bioassay and Potential Drug Screening Research Laboratory: A Research Laboratory Dedicated for Natural Products Research and Potential Drug Discovery
Proposed Grant Amount	Php 4,000,000.00
Grant Priority Area (Choose only one)	• Science, Technology, and Engineering
Duration (in months) *Maximum of 24 months*	24 months
Specific Program Offering/s to be Supported by the Grant	*BS in Biology* *BS in Environmental Science* *BS in Nutrition and Dietetics*
No. of current year's enrollees for the Program	468
No. of faculty members (full-time and part-time) under the identified program offering/s	29
Name of Project Leader	Kier P. Gasga, RChE, RChT
Contact Information *(Phone number and Email address)*	kier.gasga@parsu.edu.ph +639459734519 / +639506413783

II. OUTCOMES, GOALS, AND OBJECTIVES

Describe the project and provide the rationale or justification for it. State the expected outcomes vis-a-vis the goals and objectives.

Background of the Proposal

Research culture, and its cultivation in higher education institutions (HEIs), is of great importance. Nevertheless, universities in the developing world, including Philippines, have retained strong teaching functions and weak research functions. The paucity of research in the country has pushed Commission on Higher Education (CHED) to zealously push for a stronger research orientation among the HEIs and has provided policies and mandates that are largely geared towards the improvement of research productivity.

Nevertheless, a strong research orientation should not only be seen amongst the faculty members of universities and colleges but must also be translated to its students.

Research is an integral part of a student's life. It gives an opportunity to the student for self-learning and prepares students for future opportunities. The purpose of education is not only collecting degrees but also improve students' overall development. It is the duty of a HEIs to motivate and inspired student to adopt new approaches of research in going beyond the borders of curriculum. By doing research work, one can contribute to the overall wellbeing of the society. For this reason, it is safe to assume that a research-oriented graduate is a graduate ready for the challenges of the industry, whatever it may be.

The COVID-19 pandemic posed disruption in the higher education landscape of the country. Nevertheless, it pushed for researches that aims to find remedies in alleviating the symptoms of this virus. In the search for remedies, researchers have come back to nature. This opened a big opportunity for students taking up biology and nature and health related programs to engage in similar researches. In which, HEIs must take measure to remain relevant to this and produce competitive research-oriented graduates that could keep up with this aggressive landscape.

Partido State University (PSU) is the sole state university and premier higher education institution in the fourth district of Camarines Sur. Attracting students from the district and neighboring towns, thousands of students have enrolled to various program offerings of the university which includes BS Biology, BS Environmental Science, and BS Nutrition and Dietetics. In congruence with PSU's vision to be, among others, a leader of bio-resource science in the Bicol region, it has inspired and enabled students to engage in research and produce research outputs that could be beneficial to the community.

The university is in the heart of Partido District. Partido is the largest cluster of municipalities in the province of Camarines Sur, described as generally hilly and mountainous with flat central area known as Bikol plain. It is generally flat at the midsection between mountain ranges, with other small coastal plains on the northeastern coast. These areas are circumscribed by rolling hills, while mangroves and nipa swamps form along its coasts. The district also includes the Caramoan peninsula that covers an exotic group of islands as well as a rough and mountainous terrain with an elevation as high as 904 meters above sea level that now serves as favorite ecotourism destinations. Lastly, located within this district is the Mount Isarog Natural Park that is deemed to have high biodiversity.

PSU has seven campuses situated in municipalities within the district and are strategically located to areas where natural products could be collected. For example, Goa (where BS Biology is offered) and Tinambac Campuses (where BS Environmental Science is offered) are at proximity to Mt. Isarog, a stratovolcano rich in diversity and with various floras in its montane forest that could potentially be studied by students through the establishment of the proposed research laboratory.

With the establishment of Bioassay and Potential Drug Screening Research Laboratory, students are given a chance to specialize in researches that concerns natural products and possible drug discovery. This could usher undergraduate researches ranging from ethnobotanical and ethnopharmaceutical studies of plants, extractions and bioassays, to application and development of goods and products from which including, but not limited to, oral concoctions and pills, teas, and topical creams. Moreover, students are as well provided with a specialized platform to conduct instructional-related experiments that could provide first-hand actualization of the theoretical learnings from textbooks, hence, strengthening their capabilities as students and as industry leaders in the future.

Stipulated in CHED Memorandum Order No. 52 s. 2016 titled "Pathways to Equity, Relevance, and Advancement in Research, Innovation, and Extension in Philippine Higher Education", higher education institutions in the Philippines are encouraged to become platforms for research and development among others. With this proposal, Partido State University is geared to become a platform of research and development and a builder of pool of research-oriented graduates specializing in the screening of natural resources for possible drug discovery.

Objectives vis-à-vis Expected Outcome

Objectives	Expected Outcome
This proposal generally aims to establish a research laboratory focusing primarily on natural products, potential drug discovery, and its applications to be used by students at Partido State University. Through which, it aims to:	***Outright*** Establishment of Bioassay and Potential Drug Screening Research Laboratory.
1. Develop a natural products laboratory manual utilizing the equipment and reagents procured;	1. Copyrighted natural products laboratory manual ready for implementation and use of students. ***For 3 years afterwards***
2. Reinforce the instructional capability by providing a platform for the actualization of theoretical learnings;	2. At least (2) instructional-related experiments from target programs conducted in the laboratory.
3. Provide a training space for undergraduate students through hands-on analytical experiments and researches, hence generating research-oriented graduates; and	3. At least one (1) student in target programs trained in the laboratory through conduct of researches and/or undergraduate thesis; and

4. Contribute to the research productivity of the university by ushering students in research output dissemination.	4. At least one (1) undergraduate research conducted in the laboratory per year that published in WOS/Scopus-Indexed, CHED-recognized, or relevant internationally-refereed journal and/or at least one (1) undergraduate research conducted in the laboratory per year that presented in international, national, or regional conference/fora;

III. LOGICAL FRAMEWORK

The log frame shall provide the goals, objectives, outputs, activities, and their corresponding indicators of success of the project including the means of verification and the assumptions and risks. Risks as identified should be provided with appropriate strategies for its mitigation.

Goals/ Objectives	Activities	Outputs	Verifiable Indicators	Means of Verification	Assumptions and Risks
Establish a research laboratory focusing primarily on natural products, potential drug discovery, and its applications to be used by students at Partido State University	Procurement of necessary reagents, equipment, and glassware.	Establishment of Bioassay and Potential Drug Screening Research Laboratory.	Percent of number of reagents, equipment, and glassware received against the requested number.	Physical inventory of reagents, equipment, and glassware.	Acquisition of requested reagents, equipment, and glassware within the time period.
Develop a natural products laboratory manual utilizing the equipment and reagents procured;	Review of related literature and gathering of information for the development of a specialized laboratory manual.	Copyrighted natural products laboratory manual ready for implementation and use of students.	Copy of reviewed, piloted, and copyrighted natural products laboratory manual.	Copy of the natural products laboratory manual; and/or Certificate of Copyright/ Recordation for the laboratory manual from IPOPhl.	Completion of the laboratory manual within the given time period.
Reinforce the instructional capability by providing a platform for	Opening of the laboratory for instructional purposes	At least (2) instructional-related experiments from target	Number of instructional-related experiments conducted in	Approved request letter to conduct instructional-related	Congruence of the experiments to be conducted with the

the actualization of theoretical learnings.	through the issuance of memoranda and IECs.	programs conducted in the laboratory.	the laboratory.	experiments in the laboratory.	services of the laboratory.
Provide a training space for undergraduate students through hands-on analytical experiments and researches, hence generating research-oriented graduates.	Opening of call for undergraduate research proposals for students. To be included in the memoranda to be issued for the call for proposals is the provision of subsidy on the use of equipment and reagent available in the laboratory.	At least one (1) student in target programs trained in the laboratory through conduct of researches and/or undergraduate thesis	Number of undergraduate theses completed using the services of the laboratory.	Copy of approved undergraduate research/thesis proposal; and Copy of completed undergraduate research/thesis with the laboratory highlighted on the acknowledgement.	Congruence of the research study/thesis to be conducted with the services the laboratory.
Contribute to the research productivity of the university by ushering students in research output dissemination.	Assist students conducting research to accurate analysis and strengthen the research conducted thus increasing the chance to be accepted for publication and/or presentation.	At least one (1) undergraduate research conducted in the laboratory per year that published in WOS/Scopus-Indexed, CHED-recognized, or relevant internationally-refereed journal and/or at least one (1) undergraduate research conducted in the laboratory per year that presented in international, national, or regional conference/for a.	Number of Publication/s and/or Presentation/s of undergraduate research study/ies that utilized the services of the laboratory.	*For Publication:* Copy of Acceptance Letter and Certificate of Publication; and Copy of Published Research Article. *For Presentation:* Copy of Acceptance Letter and Certificate of Presentation; and Documentation of Presentation.	Acceptance of completed research study to publication or presentation.

IV. GANTT CHART

Gantt Chart for Year 1 (12 months)

Milestones	Activities and Tasks	YEAR 2											
		1	2	3	4	5	6	7	8	9	10	11	12
Approved PPMP and PR; Request for price quotation.	Preparation and submission of project procurement management plan (PPMP) and purchase request (PR).												
Price quotation from bidders/potential suppliers.	Call for Bidders/potential suppliers												
Notice of Award; Contract Award.	Awarding of contract to winning bidder(s) through BAC resolution												
Laboratory space ready for installation of equipment.	Retrofitting of laboratory space												
Inventory of received and installed equipment.	Receipt and physical inventory of acquired supplies and materials, and equipment including installation												
Draft laboratory manual on bioassay and potential drug discovery.	Development of natural products laboratory manual for the use of students.												
Progress Report & Fund Liquidation Report	Submission of progress report and partial liquidation report.												

Gantt Chart for Year 2 (12 months)

Milestones	Activities and Tasks	YEAR 2											
		1	2	3	4	5	6	7	8	9	10	11	12
Inventory of received and	Receipt and physical inventory of acquired												

installed equipment.	supplies and materials, and equipment including installation												
Draft laboratory manual on bioassay and potential drug discovery.	Development of natural products laboratory manual for the use of students.												
Laboratory manual on bioassay and potential drug discovery ready for pilot testing.	Evaluation of laboratory manual by internal and external reviewers.												
Establishment of Bioassay and Potential Drug Screening Research Laboratory.	Turn-over of complete requested laboratory equipment and apparatus, including installation. (Inclusive of training/tutoring of equipment use and maintenance)												
Feedback on the use of laboratory manual vis-à-vis use of laboratory equipment for improvement.	Pilot testing of laboratory manual vis-à-vis use of laboratory equipment and reagents by students.												
List of undergraduate researches to be conducted in the laboratory.	Call for research proposals for undergraduate students to be conducted in the laboratory, subsidizing reagents and equipment use.												
Final Report & Fund Liquidation Report	Submission of final report of implementation and liquidation report.												

VI. WORK AND FINANCIAL PLAN (WFP)

This component shall itemize the funding requirement per object of expenditure per month. ***Funding for Personnel Services and MOOE shall be reported as the counterpart of the proponent HEI.*** *The counterpart of the proponent HEI shall also be indicated. Please also accomplish the WFP template in Excel Format which is provided and downloadable from this link.*

PROPOSED WORK AND FINANCIAL PLAN (WFP) ***Note:** Proposed allocation for equipment, facility or systems shall be supported by a market survey containing quotations of potential suppliers including the complete specifications of each proposed equipment, facility, or system. The counterpart of the proponent HEI shall be also indicated.*					
Line Item, Budget, Particulars, and Specifications	***Unit Cost***	***Quantity***	***Duration (in months)***	***HEI Counterpart***	***Amount Requested from CHED***
Personnel Services (PS) *(Should be HEI Counterpart)*					
-	-	-	-	-	--
			SUBTOTAL PS	***0.00***	-
Maintenance and Other Operating Expenses (MOOE)					
Retrofitting of laboratory space	-	-	-	500,000.00	-
Reagents & Bioassay Kits	-	-	-	400,000.00	-
Glassware	-	-	-	100,000.00	-
			SUBTOTAL MOOE	***1,000,000.00***	-
Equipment Outlay (EO) *(includes equipment with unit cost of P15,000 or more)*					
UV-Vis Spectrophotometer *(Stray Light:<0.02%T at 340 nm, 400 nm; <1.0%T at 198 nm.* *Source: Lamp Deuterium; Tungsten-Halogen.* *Source Lamp: Life2000 hours.* *Beam Type: Double.* *Bandwidth: 1.0 nm.* *Min Wavelength (nm):190.* *Max Wavelength (nm):1100.* *Wavelength Accuracy: ±0.1 nm.)*	900,000.00	1	-	-	900,000.00
Microplate Reader *(Read Mode: Absorbance Fluorescence (top/bottom read)* *Wavelength ranges: Abs: 200 - 1000 nm* *FL Ex: 250 - 850 nm* *FL Em: 360 - 850 nm (M2)* *FL Em: 250 - 850 nm (M2e))*	800,000.00	1	-	-	800,000.00
Rotary Evaporator *(Volume of Rotary Bottle: 1L 24# Standard Aperture* *Volume of Recycling Bottle: 0.5L 24# Standard Aperture* *Vacuum Degree: 0.098Mpa* *Rotating Power: 30W* *Brushless Electric Machine* *Rotating Speed: 0-120rpm* *Heating Power: 1KW*	350,000.00	1	-	-	350,000.00

Temperature Control Range of Bath Kettle: 0-99℃ Temperature Accuracy: ±1℃)					
Analytical Balance *(Weighing Capacity:250g Scale Type: Analytical Balance Resolution: 0.0001 g Parts Counting: Yes Plug Type: Type B - North American 3-pin, Type C - European Plug, Type G - British 3-pin)*	55,000.00	2	-	-	110,000.00
Hot Plate with Magnetic Stirrer *(AC/DC input: 230 V, 50 - 60 Hz 230 V Material: material ceramic plate Feature: CE compliant stirring speed 200 - 1,300 rpm)*	55,000.00	2	-	-	110,000.00
Centrifuge *(Max. Speed: 16000rpm Max. RCF: 20600×g Max. Capacity: 6×100ml)*	150,000.00	1	-	-	150,000.00
Refrigerated Centrifuge *(Max. Speed: 16000rpm Max. RCF: 20600×g Max. Capacity: 6×100ml Temperature range: -15°C – 30°C)*	300,000.00	1	-	-	300,000.00
Shaker Incubator *(Capacity: 270L Temp Range: RT+5-65℃ Temp Precision: 0.1℃)*	100,000.00	1	-	-	100,000.00
Oven Dryer *(Temperature Range: RT-10~200℃ Temperature Uniformity: 2.5% Internal Dimension (W×D×H, mm): 280*280*350 Shelves: 5)*	200,000.00	1	-	-	200,000.00
Laminar Hood/Biosafety Cabinet Level II *(Nominal Size: 1.3 meter (4') Filtration Efficiency: ULPA: ≥99.9995%, @0.12µm Downflow Velocity: 0.35 m/s Inflow Velocity: 0.55m/s)*	400,000.00	1	-	-	400,000.00
Top-loading Autoclave *(Capacity: 60L Steam Source: Electrically generated in-chamber Operating Range: Up to 277°F, 32 psi (136°C, 2.2 Bar) Working Dimensions:13.8" (350mm) diameter x 24.4" (620mm) deep Chamber Material:Brushed grade 316L stainless steel Door Material: Burnished grade 316L stainless steel)*	400,000.00	1	-	-	400,000.00
				SUBTOTAL EO:	***0.00***	***3,820,000.00***
ADMINISTRATIVE COST					

(maximum of 5% of total amount requested from CHED)		
SUBTOTAL ADMIN COST		***180,000.00***
TOTAL AMOUNT REQUESTED FROM CHED:	***1,000,000.00***	***4,000,000.00***

VII. SUSTAINABILITY PLAN

(Describe the activities and mechanisms to be put in place to ensure the sustainability of the intervention after the end of the project.)

Financial Sustainability and Operation Continuity (Sustainability Plan)

The Bioassay and Potential Drug Screening Research Laboratory shall operationalize for the first two years under the joint funding from CHED-IDIG and the university. CHED-IDIG shall provide the funding for the acquisition of equipment whereas Partido State University shall provide the physical space in which the equipment, as well as, reagents and glassware, shall be installed and stored. Acquired instruments are expected to support and reinforce the instruction of the university and assist students in their experiments and undergraduate researches.

The maintenance of equipment and the restocking of chemical reagents shall be charged against the yearly appropriation for support to operations program. Moreover, due to the nature of the laboratory, technical services may be extended to outside clients with corresponding fee without compromising the services of the laboratory to the students.

Control and evaluation of the financial flow, particularly in areas of budgeting and accounting, shall be handled by the laboratory head. Budget received and collected fees shall be integrated to the institutionalized accounting system. In compliance to the university's accounting and auditing system, the director shall prepare and submit quarterly/biannual report to be subjected to the institution's auditing process by the internal auditor.

Staffing

The Bioassay and Potential Drug Screening Research Laboratory shall require the following personnel in order to operationalize effectively:

Laboratory Head. The laboratory head shall manage and oversee the operation and administration of the Bioassay and Potential Drug Screening Research Laboratory. Assisted by a laboratory associate, the head shall ensure that quality assurance program is established and maintained to assure quality of laboratory services, including care and maintenance of equipment. The head shall also be in charged in the development of the natural products laboratory manual. The laboratory head may be a teaching personnel and may either be a registered chemist or a registered chemical technician and shall have a deloading of 9 teaching units.

Laboratory Associate. The associate shall assist the laboratory head in the administration of laboratory. The laboratory head may be a teaching personnel with background on natural products research and development and shall have a deloading of 3 teaching units.

Laboratory Aide. The aide shall assist the labor laboratory head in clerical functions, and the like, as well as maintenance and quality assurance.

VIII. PROJECT TEAM

This component shall enumerate the project team including the terms of reference of each member. Copies of the curriculum vitae of each member shall also be attached.

(Kindly add rows as necessary.)

No.	Name of Project Team Member	Designation in HEI and Employment Status	Role in the Project
1.	Engr. Kier P. Gasga	Research and Development Coordinator; Instructor 1 (Temp)	Project Leader *(Laboratory Head)*
2.	Prof. Patricia M. Candelaria	Vice-President for Research, Extension, and Knowledge Management; Professor 4 (Perm)	Project Member *(Project Institutional Coordinator)*
3.	Diomerl Edward B. Baldo	Planning Coordinator; Instructor 1 (Temp)	Project Member *(Laboratory Associate)*

PREPARED BY:

KIER P. GASGA, RChE, RChT

PROJECT LEADER

DATE: 15 December 2021

- I hereby certify that I have reviewed and that I am endorsing the project proposal as responsive to the needs of our higher education institution.
- I hereby certify that this project proposal **has not been previously funded** by the CHED and other government agencies.
- I hereby certify that this project proposal **has not been submitted for funding** in other government agencies or other programs of CHED in the previous years.

RAUL G. BRADECINA, PhD

PRESIDENT
PARTIDO STATE UNIVERSITY

DATE: 15 December 2021